

TopAGNPS User Manual

Document Revision:

October 27, 2022

For use beginning with

TopAGNPS v6.00.a.015

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TopAGNPS User Manual

Introduction:

The purpose of this document is to; 1.) Introduce TopAGNPS and briefly describe the various modules within TopAGNPS, 2.) Describe the contents of the “TopAGNPS.CSV” control file and command line options that may be used to control the processing behavior of TopAGNPS, and 3.) Describe the inputs and outputs of TopAGNPS.

TopAGNPS is a topographic parameterization tool that uses a Digital Elevation Model (DEM) as its primary input to identify various features such as surface drainage, flow direction and flow paths. TopAGNPS uses this information to subdivide watersheds along drainage divides into sub-catchments typically referred to as “cells” and to delineate a drainage network composed of stream channels typically referred to as “reaches”. This is accomplished using the DEDNM, RASPRO, RASFOR, and AgFlow modules.

Using this information along with other user-specified input data, TopAGNPS also allows for the determination of Potential Ephemeral Gully (PEG) points using module PEG, allows the user to define riparian buffer zones using module AgBuf, define wetland areas using module AgWet, identify and produce output options for selected cells and reaches along a CONCEPTS corridor using module CONCEPTS, and identify potholes using module POTHOLE.

TopAGNPS Fortran code conforms to ANSI standard Fortran 2015.

Modules:

TopAGNPS is a suite of modules combined into one executable. The included modules are DEDNM, RASPRO, RASFOR, AGFLOW, AGBUF, AGWET, CONCEPTS, PEG, and POTHOLE. A typical scenario would be to run each of the modules in sequence as listed above. However, if not all input files are present then TopAGNPS will process as much as is possible.

The first step in processing the DEM data is accomplished by running modules DEDNM, RASPRO, RASFOR, and AgFlow. These modules will produce much information in the way of spatial ASCII raster grid files such as the watershed boundary, watershed subdivisions, drainage network, massaged elevation data, contributing drainage areas, slope information, and drainage flow paths.

The second step of processing uses the generated spatial data along with other user-supplied information to:

- 1.) Produce riparian buffers using module AgBuf.
- 2.) Produce wetland areas using module AgWet.
- 3.) Identify CONCEPTS corridors using CONCEPTS.
- 4.) Produce PEG points using module PEG.
- 5.) Identify potholes using module POTHOLE.

DEDNM, RASPRO, RASFOR, and AgFlow must be run in sequence prior to running AgBuf, AgWet, CONCEPTS, PEG, or POTHOLE. AgBuf, AgWet, CONCEPTS, PEG, and POTHOLE are independent modules that may be executed separately. However, there are two files that PEG produces that may be

optionally used as input into AgBuf and AgWet. These files are briefly listed in the “Inputs and Outputs” section of this document.

Control file: “TOPAGNPS.CSV” or “TOPAGNPS.XML”

The recommended procedure of running TopAGNPS is to use a control file. The control file is an optional file. However, it is recommended in that it allows the user to control the behavior of processing that occurs and the output files that are produced. If used, it must be called “TOPAGNPS.CSV” (comma-delimited) or “TOPAGNPS.XML” (xml-formatted) and be located in the executable path unless the command line option “/f:” is used as described in the “Command line options:” information located later in this document.

The control file, by default, must be either a comma separated value (CSV) file or an xml-formatted file. If the csv-format is used, it must contain two lines. The first line is a list of the keywords separated by a comma “,” that the user wants to use to control processing. “Keywords” will be defined later in this document. The second line is a list of corresponding values or blanks, where acceptable, also separated by a “,”. The values must be entered in the same order as the keywords. Additional lines may be included for the user’s comments but are ignored by TopAGNPS.

If the xml-format is used, it must adhere strictly to the following structure with only one tag per line, with no spaces:

```
<TOPAGNPS>
<KEYWORD ID="CELLSIZE">3</KEYWORD>
<KEYWORD ID="COLS">855</KEYWORD>
<KEYWORD ID="CSA">2.0</KEYWORD>
<KEYWORD ID="DEMEDGE">1</KEYWORD>
<KEYWORD ID="DEMPROC">0</KEYWORD>
<KEYWORD ID="DNMCNT">0</KEYWORD>
<KEYWORD ID="FILENAME">C:\AGNPS\TopAGNPS\DEDNM.ASC</KEYWORD>
<KEYWORD ID="FORMAT">0</KEYWORD>
<KEYWORD ID="MODULE">0</KEYWORD>
<KEYWORD ID="MSCL">30.0</KEYWORD>
<KEYWORD ID="KEEPFILES">0</KEYWORD>
<KEYWORD ID="NODATA">-9999.</KEYWORD>
<KEYWORD ID="OPTIMIZE">0</KEYWORD>
<KEYWORD ID="OUTCOL">464</KEYWORD>
<KEYWORD ID="OUTFORMAT">0</KEYWORD>
<KEYWORD ID="OUTPUT">0</KEYWORD>
<KEYWORD ID="OUTROW">83</KEYWORD>
<KEYWORD ID="OUTSNAP">3</KEYWORD>
<KEYWORD ID="READOUT">1</KEYWORD>
<KEYWORD ID="READPATH">C:\AGNPS\TopAGNPS\Output_Files\ASC</KEYWORD>
<KEYWORD ID="ROWS">506</KEYWORD>
<KEYWORD ID="UTME">715774</KEYWORD>
<KEYWORD ID="UTMN">4476256</KEYWORD>
<KEYWORD ID="VERBOSE">0</KEYWORD>
</TOPAGNPS>
```

Note: Only required parameters and any desired optional parameters need be present.

TopAGNPS first checks the executable path for “TOPAGNPS.CSV” and “TOPAGNPS.XML” unless the command line argument is used to invoke TopAGNPS using the “/f:” switch as described below. If both

“TOPAGNPS.CSV” and “TOPAGNPS.XML” are found, the “TOPAGNPS.CSV” control file will be processed. If neither are found, then TopAGNPS will require two input files, as historically required, called “DEDNM.INP” and “DNMCNT.INP”. “DEDNM.INP” is the input DEM with no headers and only one elevation value per line in row-major order (r1,c1; r1,c2; r1,c3...). “DNMCNT.INP” is the control file that controls processing. This file, if used, is very specific and must be followed with strict adherence.

See “Command line options:” below for information on using the command line to specify a control file with a different location or name other than “TOPAGNPS.CSV” or “TOPAGNPS.XML” as described here.

Keywords:

The following is a list of all possible keywords with a description. If a default is allowed, it will be listed as well. The keywords are listed in alphabetical order and capitalized here but that is not a requirement within “TopAGNPS.csv”. Keywords are case and order insensitive.

“CELLSIZE”

This keyword allows the user to specify the cell size, which is the length of a square raster cell of the DEM (resolution) in meters. TopAGNPS can now process input DEMs with a resolution down to 0.1 [m]. The minimum resolution with prior versions was 1.0 [m].

Optional if “FORMAT” = 0; otherwise required.

NOTE: Keywords “CELLSIZE”, “COLS”, “NODATA”, “ROWS”, “UTME”, and “UTMN” are ignored when the ASCII raster grid format is specified for input data either by default or by use of the “FORMAT” keyword with a value of “0” or blank. The values for these keywords will be extracted from the header records of the input data.

“COLS”

This keyword allows the user to specify the number of columns in the DEM input data.

Optional if “FORMAT” = 0; otherwise required.

NOTE: Keywords “CELLSIZE”, “COLS”, “NODATA”, “ROWS”, “UTME”, and “UTMN” are ignored when the ASCII raster grid format is specified for input data either by default or by use of the “FORMAT” keyword with a value of “0” or blank. The values for these keywords will be extracted from the header records of the input data.

“CSA”

This keyword allows the user to specify the Critical Source Area in hectares.

Optional; the default value of “10.0” [ha] will be assumed. (DEFAULT=10.0 [ha], if blank or keyword not used.)

“DEMEDGE”

This keyword allows the user to specify the whether to produce a warning message and continue execution –or– to produce an error message and terminate execution when the watershed boundary touches the edge of the DEM.

Optional; the default value of “0” will be assumed.

- 0 = produce warning message and continue execution - (DEFAULT=0, if blank or keyword not used.)
- 1 = produce error message and terminate execution.

“DEMPROC”

This keyword allows the user to specify the level of processing from module DEDNM.

Optional; the default value of “0” will be assumed.

- 0 = full DEM processing - (DEFAULT=0, if blank or keyword not used.)
- 1 = DEM elevation preprocessing only
- 2 = DEM elevation preprocessing and full network generation

“DNMCNT”

This keyword allows the user to specify if the original control file “DNMCNT.INP” is to be used to control processing rather than the keywords found in “TopAGNPS.csv”.

Optional; the default value of “0” will be assumed.

- 0 = do not use the dnmcnt.inp control file - (DEFAULT=0, if blank or keyword not used.)
- 1 = use the dnmcnt.inp control file, if present

“FILENAME”

This keyword allows the user to specify the input path and filename of input DEM data.

Optional; the default value of “DEDNM.ASC” will be assumed. (DEFAULT="DEDNM.ASC", if blank or keyword not used.)

“FORMAT”

This keyword allows the user to specify the format of the input DEM data. There are currently only two formats allowed.

Optional; the default value of “0” will be assumed.

- 0 = ASCII raster grid format (ASC input format) - (DEFAULT=0, if blank or keyword not used.)
- 1 = one value per record in row-major order (original “DEDNM.INP” input format)

The ASCII raster grid formatted DEM file typically contains six header lines followed by data values in a 2-D grid format (rows, columns). The six header lines contain the following header keywords followed by one or more spaces and then a value. An example is shown below:

```
Ncols 855
Nrows 506
Xllcorner 715774.397
Yllcorner 4474737.83
Cellsize 3
NODATA_value -9999
```

Another alternative format, as shown below, contains seven header lines:

```
NCOLS 855
NROWS 506
XLLCENTER 715774.397000
YLLCENTER 4474737.830000
DX 3
DY 3
NODATA_VALUE -9999
```

If this alternative format is used then “DX” and “DY”, which represent the resolution, must be the same. TopAGNPS requires square rasters.

“KEEPFILES”

This keyword allows the user to specify whether or not to keep the intermediate output files produced from DEDNM whose file size is greater than zero. If the control file “TopAGNPS.csv” is not used, the default is “1” which will keep the intermediate output files.

Optional; the default value of “0” (do not keep) will be assumed when the control file “TopAGNPS.csv” is used; “1” otherwise. (DEFAULT=0, if blank or keyword not used.)

If this keyword is used with a value of “1”, then the output location of the intermediate output files will be placed in the location associated with the “OUTPUT” keyword (see the “OUTPUT” keyword description listed below).

Based on the “OUTPUT” keyword, the intermediate files will be stored in:

OUTPUT value = 0 (default); the same location as all of the output files.

OUTPUT value = 1; the relative path of “./Output_Files/ASC” where all spatially-formatted files and projection files are stored.

“MODULE”

This optional keyword allows the user to specify which modules are to be executed.

Note: If a command line argument is included for selecting which modules are to be executed (see “Command line options” below), then this keyword is ignored. Otherwise, it will be used.

Optional; the default value of “0” (execute all modules) will be assumed. (DEFAULT=0, if blank or keyword not used.)

Options:

“0” indicates that all modules are to be executed in sequence (DEDNM, RASPRO, RASFOR, AGFLOW, PEG, and AGBUF). This value supersedes all other options listed below.

“1” indicates that only DEDNM is to be executed.

“2” indicates that only DEDNM and RASPRO are to be executed.

“3” indicates that only DEDNM, RASPRO, and RASFOR are to be executed.

“4” indicates that only AGFLOW is to be executed.

“5” indicates that only PEG is to be executed.

“6” indicates that only AGBUF is to be executed.

“7” indicates that only AGWET is to be executed.

“8” indicates that only CONCEPTS is to be executed.

“9” indicates that only POTTHOLE is to be executed.

Multiple options may be combined. For example:

“56” (modules PEG and AGBUF will be executed.)

“012” (all modules will be executed because “0” supersedes all other options.)

“MSCL”

This keyword allows the user to specify the Minimum Source Channel Length in meters.

Optional; the default value of “100.0” [m] will be assumed. (DEFAULT=100.0 [m], if blank or keyword not used.)

“NODATA”

This keyword allows the user to specify the “nodata” value for input DEM data.

Optional if “FORMAT” = 0; otherwise required.

NOTE: Keywords “CELLSIZE”, “COLS”, “NODATA”, “ROWS”, “UTME”, and “UTMN” are ignored when the ASCII raster grid format is specified for input data either by default or by use of the “FORMAT” keyword with a value of “0” or blank. The values for these keywords will be extracted from the header records of the input data.

“OPTIMIZE”

This optional keyword allows the user to specify the method of internal loop processing. Historically, modules DEDNM, RASPRO, and RASFOR processed DEM data in row-major order. This means that the DEM data is processed by row first and columns second (r1,c1; r1,c2; r1,c3; r2,c1; r2,c2; r2,c3; ...). FORTRAN execution is faster, however, when loops are processed in column-major order (c1,r1; c1,r2; c1,r3; c2,r1; c2,r2; c2,r3; ...).

The resulting output files from these modules will differ somewhat as a result of the different loop-processing methods.

Optional; the default value of “0” (legacy row-major loop processing) will be assumed. (DEFAULT=0, if blank or keyword not used; 1= optimize execution using column-major loop processing).

“OUTCOL”

This keyword allows the user to specify the outlet column or UTM “x” location to use for the channel network.

Optional if “DEMPROC” = 1 or 2; required if “DEMPROC” = 0.

NOTE: “OUTFORMAT” option “1” must be used for UTM coordinates.

“OUTFORMAT”

This keyword allows the user to specify if the outlet location is specified by row and column coordinates or by UTM coordinates.

Optional; the default value of “0” will be assumed.

0 = the outlet is specified by row and column - (DEFAULT=0, if blank or keyword not used.)

1 = the outlet is specified by UTM x & y coordinates.

NOTE: If UTM coordinates are used then the UTM “y” coordinate must be placed in the “OUTROW” keyword column and the UTM “x” coordinate must be placed in the “OUTCOL” column.

TopAGNPS converts all UTM coordinates to row and column for processing. The conversion of the UTM coordinates here may result in the outlet location not falling exactly on the reach raster. However, the use of “OUTSNAP” either by default or specified by the user will attempt to snap the outlet to the reach raster. See the description for “OUTSNAP” below for more information.

“OUTPUT”

This optional keyword allows the user to specify whether all output files produced will be in a single folder or in an internally defined folder structure. If “OUTPUT” is used with a value of “0”, or the value is left blank, then the output files will be written to the following relative path:

./Output_Files/ASC where all spatially-formatted files and projection files are stored.

./Output_Files/CSV where all csv-formatted table files are stored.

./Output_Files/DAT where all text-formatted table files are stored.

./Output_Files/LOG where all module log and report files are stored.

If “OUTPUT” is used with a value of “1”, or the value is left blank, then all of the output files will be written to folder from which TopAGNPS was invoked.

With either option, the TopAGNPS log, warning, error, and status files are written to folder from which TopAGNPS was invoked.

Optional; the default value of “1” (all output files are written to folder from which TopAGNPS was invoked) will be assumed. (DEFAULT=1, if blank or keyword not used; 1= all of the output files will be written to folder from which TopAGNPS was invoked).

“OUTROW”

This keyword allows the user to specify the outlet row or UTM “y” location to use for the channel network.

Optional if “DEMPROC” = 1 or 2; required if “DEMPROC” = 0.

NOTE: “OUTFORMAT” option “1” must be used for UTM coordinates.

“OUTSNAP”

This keyword allows the user to specify the number of outwardly expanding passes to make searching for a substitute outlet location. This is helpful in the event that the “OUTROW” and “OUTCOL” location is not valid because, 1.) It does not fall on a channel network; 2.) It falls on the edge of the DEM; or 3.) It is a junction cell. Rather than terminating execution with an error message, this keyword instructs the code to search adjacent rasters for a valid watershed outlet location.

Optional; the default value of “2” will be assumed.
(DEFAULT=2, if blank or keyword not used; 0 = no snap)

If no valid outlet location can be found within the “OUTSNAP” number of passes, then an error message is logged and execution terminates. If a valid outlet location is found, then a message is logged containing the new outlet location and processing continues.

Searching stops with the location of a valid outlet location. In the event there are more than one valid outlet locations available within a pass, then the one with the largest uparea value is selected. In the event of a tie, the first encountered is selected.

“READOUT”

This keyword allows the user to specify that the “intermediate” output files produced from a previous run will be used as input rather than TopAGNPS having to regenerate them.

Optional; the default value of “0” will be assumed.
0 = do not use intermediate files - (DEFAULT=0, if blank or keyword not used.)
1 = use intermediate output files from previous run as input

If this parameter is used and has a value of “1”, then the “KEEPFILES” keyword option will be assumed with a value of “1” meaning that the intermediate files that are produced will not be deleted at the end of execution.

The location of the intermediate files as produced by TopAGNPS is determined by the “OUTPUT” keyword (see the “OUTPUT” keyword description above).

The intermediate files required for use with this option are:

FLOPAT.OUT, FLOVEC.OUT, INDTAR.OUT, NTGCOD.OUT, NTGCSA.OUT, NTGSCL.OUT, RELIEF.OUT, and UPAREA.OUT.

NOTE: Care must be taken when using this option in that the user input values used to produce the .OUT files from the previous run of TopAGNPS must be used when using them as input into a new run. Parameters such as number of rows and columns, cell size, critical source area (CSA) and minimum source channel length (MSCL) must match.

“READPATH”

This optional keyword allows the user to specify the location to the “intermediate” output files produced from a previous run that will be used as input to TopAGNPS.

The default path will be the path associated with the “OUTPUT” keyword (see the “OUTPUT” keyword description above).

“ROWS”

This keyword allows the user to specify the number of rows in the DEM input data.

Optional if “FORMAT” = 0; otherwise required.

NOTE: Keywords “CELLSIZE”, “COLS”, “NODATA”, “ROWS”, “UTME”, and “UTMN” are ignored when the ASCII raster grid format is specified for input data either by default or by use of the “FORMAT” keyword with a value of “0” or blank. The values for these keywords will be extracted from the header records of the input data.

“UTME”

This keyword allows the user to specify the UTM Easting coordinate of the upper left corner of the DEM.

Optional if “FORMAT” = 0; otherwise required.

NOTE: Keywords “CELLSIZE”, “COLS”, “NODATA”, “ROWS”, “UTME”, and “UTMN” are ignored when the ASCII raster grid format is specified for input data either by default or by use of the “FORMAT” keyword with a value of “0” or blank. The values for these keywords will be extracted from the header records of the input data.

“UTMN”

This keyword allows the user to specify the UTM Northing coordinate of the upper left corner of the DEM.

Optional if “FORMAT” = 0; otherwise required.

NOTE: Keywords “CELLSIZE”, “COLS”, “NODATA”, “ROWS”, “UTME”, and “UTMN” are ignored when the ASCII raster grid format is specified for input data either by default or by use of the “FORMAT” keyword with a value of “0” or blank. The values for these keywords will be extracted from the header records of the input data.

“VERBOSE”

This keyword allows the user to request extended log messages produced during processing to be included in the log file.

Optional; the default value of “0” will be assumed.

0 = produce more messages - (DEFAULT=0, if blank or keyword not used.)

1 = produce less messages

Examples:

Example 1: Shows use of selected keywords from a text editor view.

FILENAME, FORMAT, DEMPROC, OUTROW, OUTCOL, CSA, MSCL, UTME, UTMN, ROWS, COLS, CELLSIZE, NODATA, OUTSNAP, DNMCNT
DEDNM.ASC, 0, 0, 85, 461, 2, 30, 715774, 4476256, 506, 855, 3, -9999, 3, 0

Example 2: Shows use of selected keywords from a spreadsheet view.

Note: Keywords left out of this example either have default values or required information will be extracted from the header lines of the input DEM data. For example, the default format is “0” as is the default for DEM processing in the absence of the keywords “FORMAT” and “DEMPROC”. The number of rows and columns, the UTM coordinates, the cell size, and the nodata value will be extracted from “DEDNM.ASC”. Also, header information within the DEM data will override user-supplied values in “TopAGNPS.csv”.

FILENAME	OUTROW	OUTCOL	CSA	MSCL	OUTSNAP
DEDNM.ASC	83	464	2	30	3

Additional notes:

- 1.) To preserve backward compatibility:
 - a.) If this control file does not exist in the executable path, then "DNMCNT.INP" and "DEDNM.INP" must exist for execution to occur. This means that execution will occur as historically done.
 - b.) If this control file does exist, then it is processed first and the input file is read in per the specified format and processed.
- 2.) If "FORMAT" option is set to 0 (ASC) and "DNMCNT" is set to "0" or blank, then the following keywords are optional: "CELLSIZE", "COLS", "ROWS", "UTME", "UTMN" and "NODATA". Information for these

will be extracted from the header records within the DEM "FILENAME". If "CSA" and "MSCL" keywords are not supplied their defaults will be used. Additionally, "OUTROW" and "OUTCOL" keywords are required only if "DEMPROC" = 0.

3.) If "DNMCNT" is set to "1":

a.) If the user sets the "DNMCNT" value to "1" and the "dnmcnt.inp" file exists then all keywords in "TopAGNPS.CSV" with related values in the "dnmcnt.inp" file will be ignored. For example, the number of rows and columns are specified in the "dnmcnt.inp" file. Therefore, if the keywords "ROWS" and "COLS" are included in "TOPAGNPS.CSV", they will be ignored.

4.) "OUTSNAP" is used by default with a value of "2". When the user-selected outlet is not a valid outlet, the code searches the adjacent rasters in an outwardly expanding direction for the number of passes defined using this keyword or its default "2".

The search proceeds from the upper-left corner of the search window and proceeds by incrementing the column count first and then subsequent rows. All valid rasters are considered and the raster with the largest uparea value is selected. If multiple valid rasters have the same largest uparea value then the first raster encountered with that value is selected.

If no valid substitute raster can be found within the search criteria then an error message is produced and execution terminates.

The following tables give an example of searching order for 3 different numbers of "OUTSNAP" passes. The allowable maximum limit is the lesser of the rows or columns in the DEM. Note: The grayed center raster of each table represents the user-specified outlet location.

"OUTSNAP" = 1

1	2	3
4	5	6
7	8	9

"OUTSNAP" = 2

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25

"OUTSNAP" = 3

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31	32	33	34	35
36	37	38	39	40	41	42
43	44	45	46	47	48	49

Command line options:

TopAGNPS may be invoked with up to two arguments; one to control which modules are to be executed and the other to specify an alternate control file location, name, and possible format. One way the argument may be included is through a command line execution of TopAGNPS in the following form:

TopAGNPS.exe /01234567 where:

“/” is the switch designator which informs TopAGNPS that an argument is included. There must be a space between “TopAGNPS.exe” and the “/” character but no space between the “/” and the remainder of the argument.

“0” indicates that all modules are to be executed in sequence (DEDNM, RASPRO, RASFOR, AGFLOW, PEG, and AGBUF). This value supersedes all other options listed below.

“1” indicates that only DEDNM is to be executed.

“2” indicates that only DEDNM and RASPRO are to be executed.

“3” indicates that only DEDNM, RASPRO, and RASFOR are to be executed.

“4” indicates that only AGFLOW is to be executed.

“5” indicates that only PEG is to be executed.

“6” indicates that only AGBUF is to be executed.

“7” indicates that only AGWET is to be executed.

“8” indicates that only CONCEPTS is to be executed.

“9” indicates that only POTHOLE is to be executed.

Note that each module requires input files and produces output files; many of which are inputs into the next module in sequence. For example, PEG, AGBUF, AGWET, CONCEPTS, and POTHOLE modules require, as input, files produced by DEDNM, RASPRO, RASFOR, and AGFLOW. Therefore, care must be exercised when invoking TopAGNPS in this manner.

The other argument is in the form of:

TopAGNPS.exe /f:filename where:

“/” is the switch designator which informs TopAGNPS that an argument is included. There must be a space between “TopAGNPS.exe” and the “/” character but no space between the “:” and the remainder of the argument.

“f:” indicates that the remainder of the argument is the path and filename of the control file. The control file, as described above, must be either a csv-formatted file with a “CSV” extension or an xml-formatted file with a “XML” extension.

For example:

TopAGNPS.exe /f:/Parent_folder/My_Control_File.CSV

Or

TopAGNPS.exe /f:/Parent_folder/My_Control_File.XML

XML-formatted control file:

TopAGNPS does allow for a very strict but simple XML input format where the user can specify one or more of the keywords as described earlier in this document. It takes the following form:

```
<TOPAGNPS>
<KEYWORD ID="CELLSIZE">3</KEYWORD>
<KEYWORD ID="COLS">855</KEYWORD>
<KEYWORD ID="CSA">2.0</KEYWORD>
<KEYWORD ID="DEMEDGE">1</KEYWORD>
<KEYWORD ID="DEMPROC">0</KEYWORD>
<KEYWORD ID="DNMCNT">0</KEYWORD>
<KEYWORD ID="FILENAME">C:\AGNPS\TopAGNPS\DEDNM.ASC</KEYWORD>
<KEYWORD ID="FORMAT">0</KEYWORD>
<KEYWORD ID="KEEPFILES">0</KEYWORD>
<KEYWORD ID="MSCL">30.0</KEYWORD>
<KEYWORD ID="MODULE">0</KEYWORD>
<KEYWORD ID="NODATA">-9999.</KEYWORD>
<KEYWORD ID="OPTIMIZE">0</KEYWORD>
<KEYWORD ID="OUTCOL">464</KEYWORD>
<KEYWORD ID="OUTFORMAT">0</KEYWORD>
<KEYWORD ID="OUTPUT">0</KEYWORD>
<KEYWORD ID="OUTROW">83</KEYWORD>
<KEYWORD ID="OUTSNAP">3</KEYWORD>
<KEYWORD ID="READOUT">1</KEYWORD>
<KEYWORD ID="READPATH">C:\AGNPS\TopAGNPS\Output_Files\ASC</KEYWORD>
<KEYWORD ID="ROWS">506</KEYWORD>
<KEYWORD ID="UTME">715774</KEYWORD>
<KEYWORD ID="UTMN">4476256</KEYWORD>
<KEYWORD ID="VERBOSE">0</KEYWORD>
</TOPAGNPS>
```

Only one tag can be used per line, with no spaces, and the extension for this file must be “.XML”. It may be used in one of two ways:

- 1.) command line argument - `TopAGNPS.exe /f:My_xml_control_file.xml`
- 2.) direct control file input

Examples:

Example 1: Shows invocation of TopAGNPS with no argument and “/0” argument.

`TopAGNPS.exe` (no argument is used meaning all modules will be executed in sequence as long as the required input files are available.)

`TopAGNPS.exe /0` (argument is used; all modules will be executed in sequence as long as the required input files are available.)

Example 2: Shows invocation of TopAGNPS with one argument.

TopAGNPS.exe /3 (argument is used; modules DEDNM, RASPRO, and RASFOR will be executed.)

TopAGNPS.exe /456 (argument is used; modules AGFLOW, PEG and AGBUF will be executed.)

TopAGNPS.exe /012 (argument is used; all modules will be executed because “0” supersedes all other options.)

Example 3: Shows invocation of TopAGNPS with two arguments.

TopAGNPS.exe /1234 /f:my_csv_control_file.csv

TopAGNPS.exe /0 /f:my_xml_control_file.xml

GIS Projection Files:

GIS projection files are optional. They may be included in the following ways:

- 1.) A projection file that accompanies the input DEM file will be propagated through all of the ASCII raster grid output files. The expected filename is the same as the input DEM but with a “.prj” extension. For example, if the input DEM is named “My_DEM.asc” then the projection file needs to be named “My_DEM.prj”. Then as the full run of TopAGNPS is executed, every ASCII raster grid output file “*.asc” will have an associated projection file “*.prj” with the same contents as “My_DEM.prj”.
- 2.) Modules AgFlow, AgBuf, AgWet, CONCEPTS, PEG, and POTHOLE all require several input grid layers that are produced by DEDNM. One of those files is called “FLOVEC.ASC”. If this file has an associated projection file “FLOVEC.PRJ” then independent runs of AgBuf, AgWet, CONCEPTS, PEG, and Pothole will read this input projection file and output projection files for all of the resulting ASCII raster grid output files. For example, if “FLOVEC.PRJ” exists and only module AgBuf is executed, then all of the ASCII raster grid output files “*.asc” produced from AgBuf will have an associated projection file “*.prj”.

Inputs and Outputs:

Module input and output files:

Each module requires certain input files in order to successfully execute. The following is a list of required input files for each module and the output files from that module that is either required or optional for one of the subsequent modules. Not all output files that are produced are listed here. Also, if executing TopAGNPS on a UNIX/Linux system (case sensitive), the following list is in the expected case for each file.

DEDNM, RASPRO, RASFOR, AgFlow:

Input: Required: DEM data and control information either from “TOPAGNPS.CSV” or “DNMCNT.INP”

Output: BOUND.ASC, DEPFLT.ASC, FLOVEC.ASC, AgFlow_LS_Factor.asc, NETW.ASC, RELIEF.ASC, TSLOPE.ASC, UPAREA.ASC

AGFLOW:

Input: Required: FLOVEC.ASC, FVSLOP.ASC, HSLOPE.ASC, RELIEF.ASC, TSLOPE.ASC, SUBWTA.ASC, TASPEC.ASC, TSLOPE.ASC, and UPAREA.ASC.
Optional: AGFLOW.CSV or AGFLOW.XML

Output: AnnAGNPS_Cell_IDs.asc, AnnAGNPS_Reach_IDs.asc, AgFlow_Reach_Data.csv

PEG:

Input: Required: PEG.CSV or PEG.XML, BOUND.ASC, FLOVEC.ASC, NETW.ASC, RELIEF.ASC, TSLOPE.ASC, UPAREA.ASC, AnnAGNPS_Reach_IDs.asc, AnnAGNPS_Cell_IDs.asc.

Optional: user-specified csv-formatted PEG points file.

Output: PEG_Summary.csv, AnnAGNPS_Ephemeral_Gully_Data_Section.csv

AgBuf:

Input: Required: AGBUF.CSV or AGBUF.XML, FLOVEC.ASC, NETW.ASC, TSLOPE.ASC, UPAREA.ASC, AnnAGNPS_Reach_IDs.asc, AnnAGNPS_Cell_IDs.asc, user-specified spatial buffer zone and vegetation files.

Optional: PEG_Summary.csv and AnnAGNPS_Ephemeral_Gully_Data_Section.csv

Output: AnnAGNPS_Ephemeral_Gully_Data_Section_Revised_by_AgBuf.csv.

AgWet:

Input: Required: AGWET.CSV or AGWET.XML, BOUND.ASC, FLOVEC.ASC, RELIEF.ASC, TSLOPE.ASC, UPAREA.ASC, AgFlow_LS_Factor.asc, AnnAGNPS_Reach_IDs.asc, AnnAGNPS_Cell_IDs.asc,
Optional: PEG_Summary.csv and either AnnAGNPS_Ephemeral_Gully_Data_Section.csv or AnnAGNPS_Ephemeral_Gully_Data_Section_Revised_by_AgBuf.csv, user-specified spatial buffer zone file, vegetation file, buffer ids file, and barrier heights file.

Output: None for subsequent modules.

CONCEPTS:

Input: Required: CONCEPTS.CSV or CONCEPTS.XML, FLOVEC.ASC, AnnAGNPS_Reach_IDs.asc, AnnAGNPS_Cell_IDs.asc, AgFlow_Reach_Data.csv

Output: None for subsequent modules.

POTHOLE:

Input: Required: POTHOLE.CSV or POTHOLE.XML, BOUND.ASC, DEPFLT.ASC, FLOVEC.ASC, UPAREA.ASC, AnnAGNPS_Reach_IDs.asc

Output: None for subsequent modules.

Output folder structure:

TopAGNPS separates the output files into the following folder structure if the “OUTPUT” keyword is used with a value of “0”:

./Output_Files/ASC – This folder contains all spatial files.

./Output_Files/CSV – This folder contains all csv-formatted files.

./Output_Files/DAT – This folder contains all other text-formatted output files.

./Output_Files/LOG – This folder contains all of the various log files.

Otherwise, all output files are written to the folder from which TopAGNPS was invoked.

Other Related Documentation:

AGBUF User Manual

Please review the document called “AGBUF_User_Manual.pdf” for details on the specific use of this module.

AGFLOW User Manual

Please review the document called “AGFLOW_User_Manual.pdf” for details on the specific use of this module.

AGWET User Manual

Please review the document called “AGWET_User_Manual.pdf” for details on the specific use of this module.

PEG User Manual

Please review the document called “PEG_User_Manual.pdf” for details on the specific use of this module.

CONCEPTS User Manual

Please review the document called “CONCEPTS_User_Manual.pdf” for details on the specific use of this module.

POTHOLE User Manual

Please review the document called “POTHOLE_User_Manual.pdf” for details on the specific use of this module.